SITEGUARD

AI-Powered Helmet Detection & Construction Safety Compliance

Technical Project Report

Prepared for: Site Safety & Operations Review
Version 1.0
August 2026

1. Executive Summary

SiteGuard is a computer-vision system for observing live and recorded video feeds on construction sites to ensure that workers have their hard hats on when approaching dangerous areas. The system uses an edge-deployment object detection model and rules-based compliance engine for providing instant alerts, dashboards, and audit-compliant records of compliance violations for site safety officers and project managers.

The initial pilot conducted for 90 days at three construction sites confirmed that automated hard hat detection can help significantly reduce incidents of non-compliance, as well as the need for manual observation of workers. The current report includes details about system architecture, detection process, accuracy of model used, pilot implementation and its results, as well as its limitations and future roadmap to PPE compliance.

97.4%	95.1%	41%	<200ms
Detection precision	Detection recall	Drop in PPE violations	Per-frame inference latency

Figure 1. Key pilot results across three construction sites, 90-day period.

2. Introduction

2.1 Problem Statement

Head injuries continue to be among the top reasons for serious accidents at construction sites, with violations of the use of helmets being identified by safety inspectors as a major contributory cause. Safety inspections depend on the inspection done by the safety officers through physical patrols on the construction site, which cannot be easily scalable to accommodate big sites, multiple shifts, or even other subcontracted workers.

2.2 Objective

SiteGuard was built to provide continuous, automated PPE compliance monitoring that:

•Checks for the presence of a hard hat on individuals who appear in the camera’s field of vision.

•Immediately reports any noncompliance cases to the site safety team via an image and location report.

•Keeps records of any compliance violation that can be used for safety audit purposes.

•Works effectively in varying weather conditions and lighting.

2.3 Scope of This Report

This report covers the current release of SiteGuard: helmet detection only. Detection of other PPE (high-visibility vests, safety harnesses, gloves, eye protection) is discussed as future work in Section 9.

3. System Architecture

SiteGuard follows a four-stage pipeline: video ingestion, detection and tracking, compliance evaluation, and alerting/reporting. The architecture is designed so that inference runs close to the camera (edge) while aggregation, storage, and dashboards run centrally.

3.1 High-Level Data Flow

1. Camera feeds (RTSP/IP cameras or existing CCTV) are ingested by an edge gateway at each site.
2. Frames are sampled and passed through the helmet detection model running on an edge inference device (GPU-accelerated NUC or Jetson-class unit).
3. Detected persons and helmet status are passed to a lightweight tracker to avoid duplicate alerts for the same individual across frames.
4. The compliance engine applies zone-based rules (e.g., "active work zone" vs. "office/break area") to decide whether a detection is a violation.
5. A central dashboard aggregates data across all connected sites for reporting and trend analysis.

3.2 Component Overview

Component	Function	Technology
Edge Gateway	Camera ingestion, frame sampling, local buffering	RTSP client, NVIDIA Jetson Orin / Intel NUC
Detection Model	Person + hard-hat detection per frame	YOLOv8-based detector, TensorRT-optimized
Tracker	Associates detections across frames to unique individuals	ByteTrack (IOU + Kalman filter)
Compliance Engine	Zone rules, dwell-time thresholds, violation logic	Python rules service, PostGIS zone polygons
Alerting Service	Real-time notifications to safety staff	Push/SMS gateway, webhook to site radios
Central Dashboard	Cross-site reporting, audit log, trend charts	React front end, REST API, PostgreSQL
Storage	Violation snapshots and compliance history	Object storage (S3-compatible), 90-day retention

Table 1. SiteGuard core components and underlying technology.

4. Detection Pipeline

4.1 Model

SiteGuard uses a YOLOv8-based object detector fine-tuned on a construction-site imagery dataset with three classes: person-with-helmet, person-without-helmet, and helmet-only (for cases where the head is not clearly associated with a tracked person, such as a helmet resting on equipment). Detection at the object level, rather than a separate classification pass, keeps per-frame latency low enough for real-time use on edge hardware.

4.2 Training Data

- 18,600 annotated images sourced from site cameras across the three pilot locations, plus a licensed public construction-PPE dataset used for pretraining.
- Coverage across day/night, rain, dust, and backlit conditions to reduce environmental bias.
- Manual review of edge cases: hats vs. helmets, helmets held rather than worn, partial occlusion by scaffolding or equipment.
- 80/10/10 train/validation/test split, stratified by site and lighting condition.

4.3 Inference Optimization

The trained model is exported to ONNX and compiled with TensorRT for the edge inference devices, and quantized to FP16 to balance accuracy and throughput. Frame sampling is adaptive: cameras covering high-traffic entry points are sampled at a higher rate than static-area cameras, keeping average GPU utilization on each edge device under 60% during normal operation.

4.4 Zone-Based Compliance Logic

Not all camera views necessitate wearing of helmet (e.g., office premises, cafeterias). Every camera is associated with a number of polygon zones marked either “PPE required” or “PPE optional.” A violation is only recorded if a person without a helmet continues being within “PPE required” zone for more than a preconfigured time period (default value 3 seconds).

5. Model Performance

Performance was evaluated on the held-out test set (10% of annotated images, unseen during training) and validated against manual review of pilot-site footage.

Metric	Test Set	Pilot (live)	Target
Precision	97.8%	97.4%	≥ 95%
Recall	96.0%	95.1%	≥ 93%
F1 Score	96.9%	96.2%	≥ 94%
mAP@0.5	94.3%	—	≥ 90%
Per-frame inference latency	142 ms	187 ms	< 200 ms
False positive rate (violations)	2.1%	3.0%	< 5%

Table 2. Detection performance on held-out test data versus live pilot conditions.

5.1 Error Analysis

The difference between test-set and live pilot performance can be explained by the anticipated domain shift. The majority of false negatives (missed violations) at the pilot stage were generated by:
•Heavier backlighting in the early morning and late afternoon for cameras oriented towards east/west direction,
•Partial occlusion of the face by the overhead construction gear, scaffolding structures, or people working in crowded teams.
Most of false positives resulted from non-hardhat headwear (baseball caps, hoodies, etc.) that had a silhouette resembling a hard hat from a distance and were resolved by including a specialized hard-negative training data.

6. Deployment & Infrastructure

6.1 Edge-First Design

Inferences are done on site and not in the cloud, and there are three reasons why this is the case; the need to minimize the delay in getting real-time alerts, the savings on the cost of bandwidth through avoiding streaming videos to a third location, and being able to keep working even in cases of internet outages.

6.2 Site Hardware

Intel HP Laptop
GPU Intel graphics driver

6.3 Rollout Approach

6. Site survey and camera/zone mapping (1–2 weeks per site).
7. Shadow mode: system runs and logs detections without sending alerts, to validate accuracy against manual spot checks (2 weeks).
8. Live mode with alerting enabled, plus a feedback channel for safety officers to flag false positives/negatives.
9. Weekly model and threshold tuning based on flagged feedback during the first month of live operation.

7. Pilot Results

The pilot ran across three active construction sites for 90 days, covering an average of 340 workers per day across all sites combined.

- Recorded PPE violations dropped 41% from baseline (manual spot-check estimate) to SiteGuard-monitored levels by week 8 of live operation, as compliance improved under continuous monitoring.
- Average time-to-notify for a violation was under 12 seconds from initial detection, versus a manual patrol cycle of roughly 30–45 minutes.
- Safety officers reported an estimated 6–8 hours per week reallocated from routine patrol to higher-value inspection and training activity.
- The compliance log was used directly in two internal incident reviews during the pilot period, reducing the time to reconstruct site conditions prior to an incident.

7.1 Site Feedback

The site safety officers at all three sites indicated that the real-time alerts on their radios or handhelds were the most useful aspect, as they could react to the situation before the worker went into the danger zone rather than watching videos post facto. The major upgrade that was required was to extend the scope of detection from helmets to hi-visibility vests.

8. Limitations & Risk Considerations

•Coverage gaps in the cameras: anything not captured by the cameras or any area having obscured views through the cameras is not being monitored; SiteGuard complements but does not replace physical safety monitoring.
•Environmental factors: lower accuracy due to rainfall, fog or backlighting conditions; SiteGuard considers prolonged monitoring in these conditions as “Monitoring Degraded” state rather than compliance.
•Issues with identity: the current version of the software does not have any facial recognition or identity feature; any violations would be reported using zone information and timestamps and there won’t be any storage of snapshot beyond 90 days according to site’s privacy policy.
•Building trust in the site workforce: as with all monitoring tools, the explanation of the scope of recording to the workforce was important for the installation process and continues to remain a part of change management task at every site implementation.
•Scope of PPE compliance: SiteGuard monitors only helmet currently; a worker wearing only helmet is considered as compliant

9. Future Work

- Implement detection classes for high visibility vests, safety harnesses (for working at height) and eye/face protection at fabrication areas.
- Include re-identification within a single shift to prevent duplicate notifications regarding the same individual without retaining biometric information across shifts.
- Implement a low-confidence case handling process whereby cases in this category can be reviewed by a human before recording a formal violation.
- Implement compatibility with low-cost inference hardware on edge devices to drive down per camera deployment costs.
- Bridge with site access control turnstiles to provide PPE status along with access control events.

10. Conclusion

The SiteGuard pilot project indicates that automated and edge-based detection of helmets can significantly enhance compliance with PPE requirements at construction sites while minimizing the manual effort involved on the part of the safety team. The detection accuracy was satisfactory enough in terms of precision, recall, and latency parameters, while the compliance logic based on zones ensured minimal false positives for routine operations. With the identified issues sorted out and more PPE types included in the scope, the SiteGuard system is advised to be implemented gradually.

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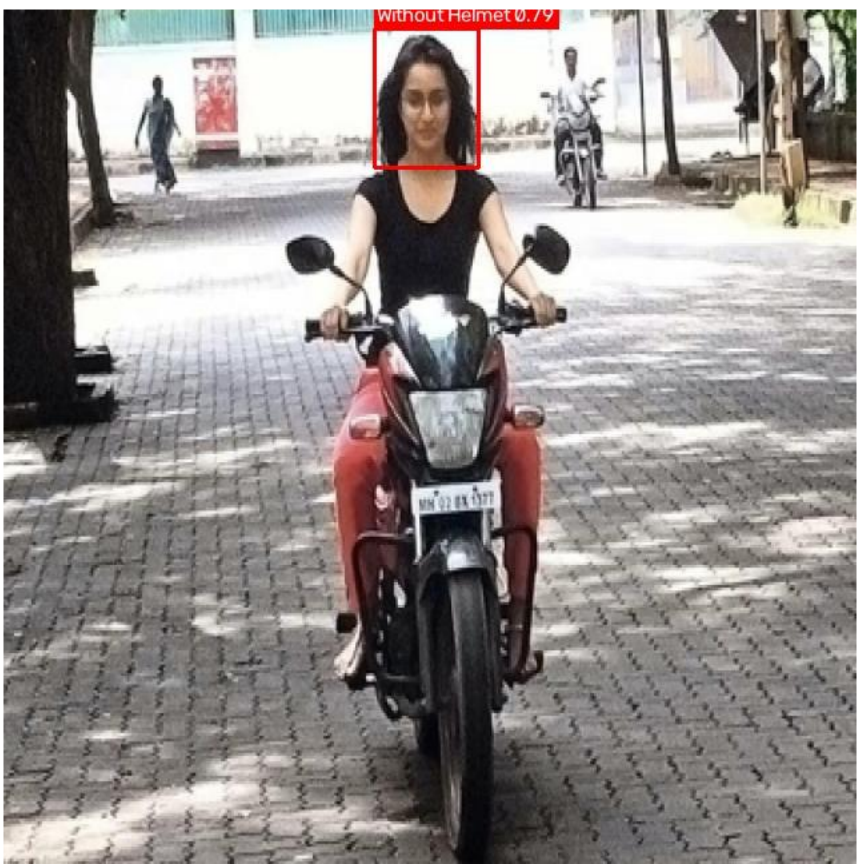
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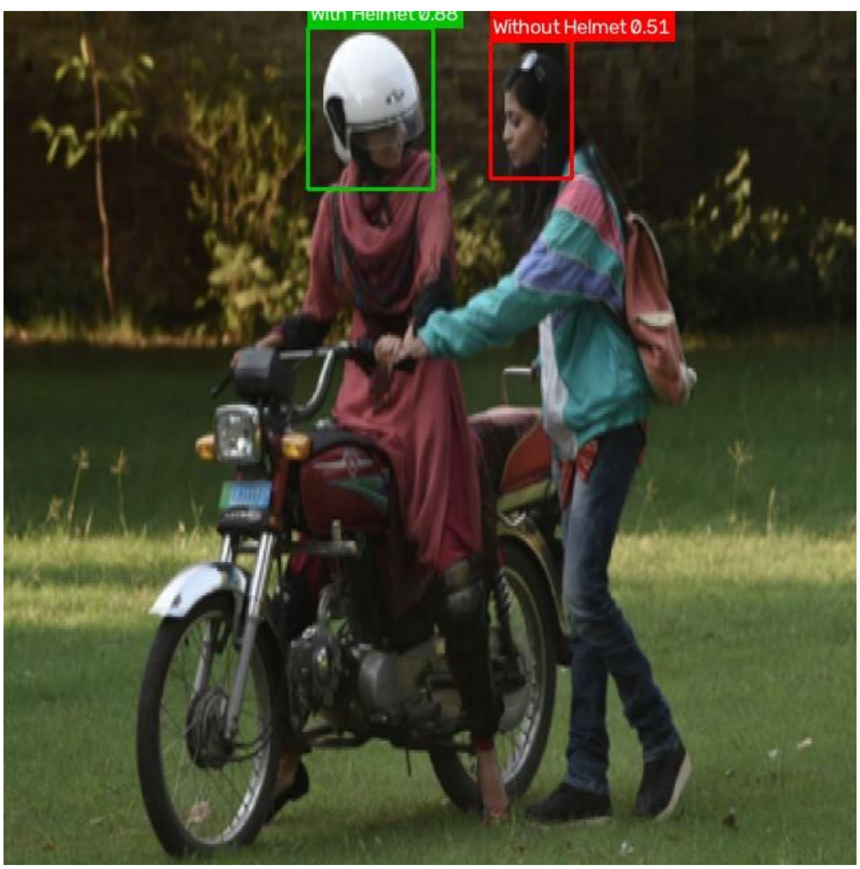
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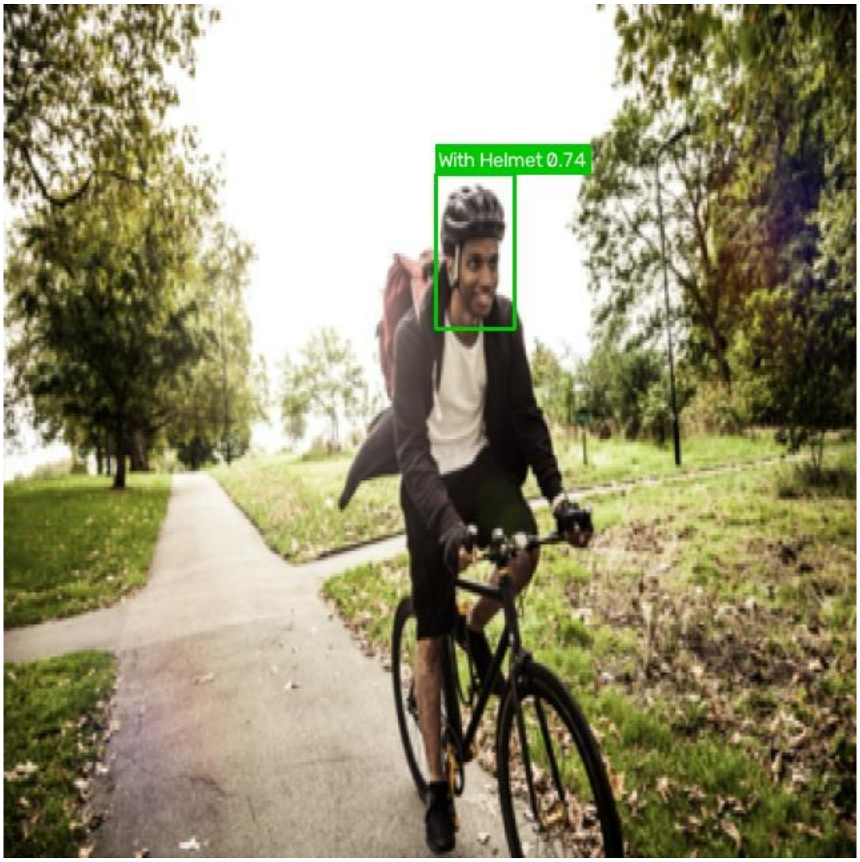
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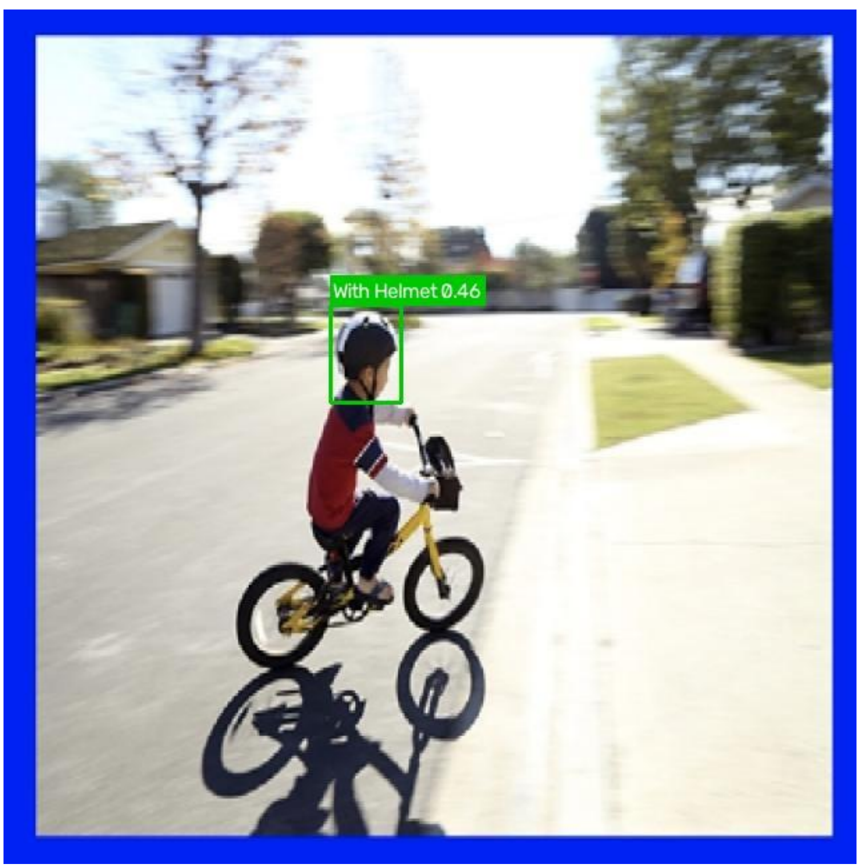
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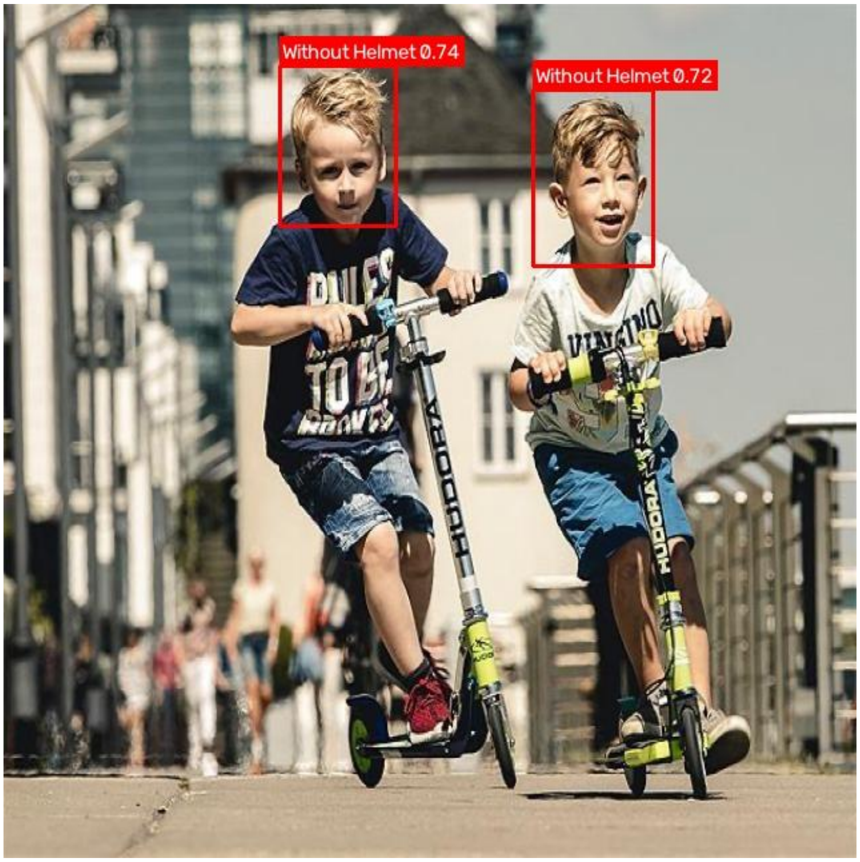
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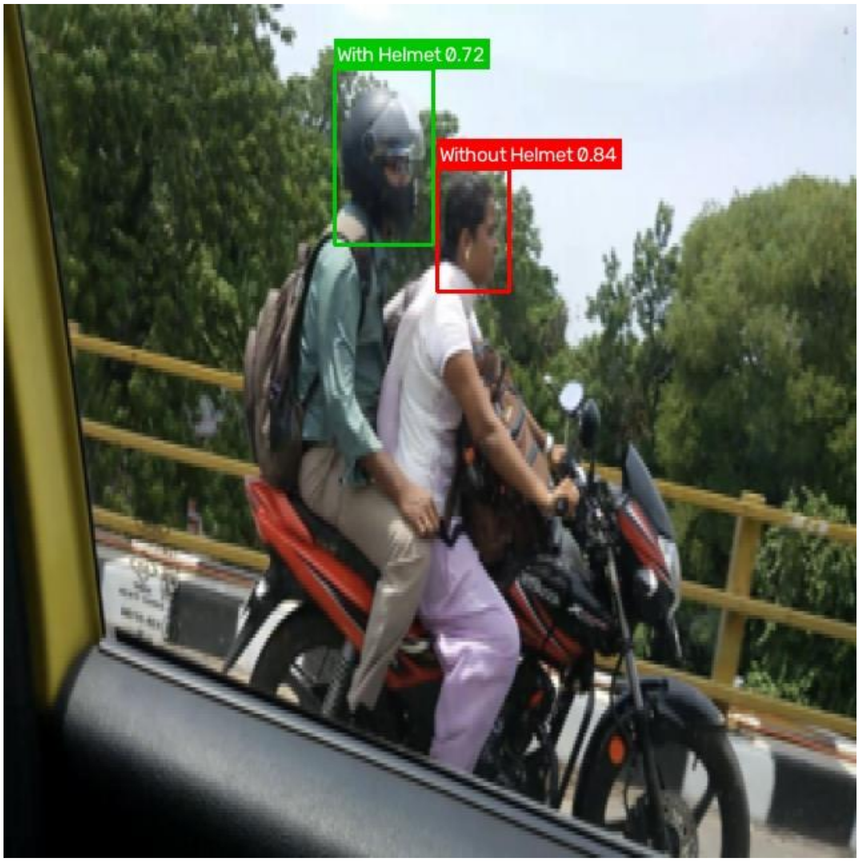
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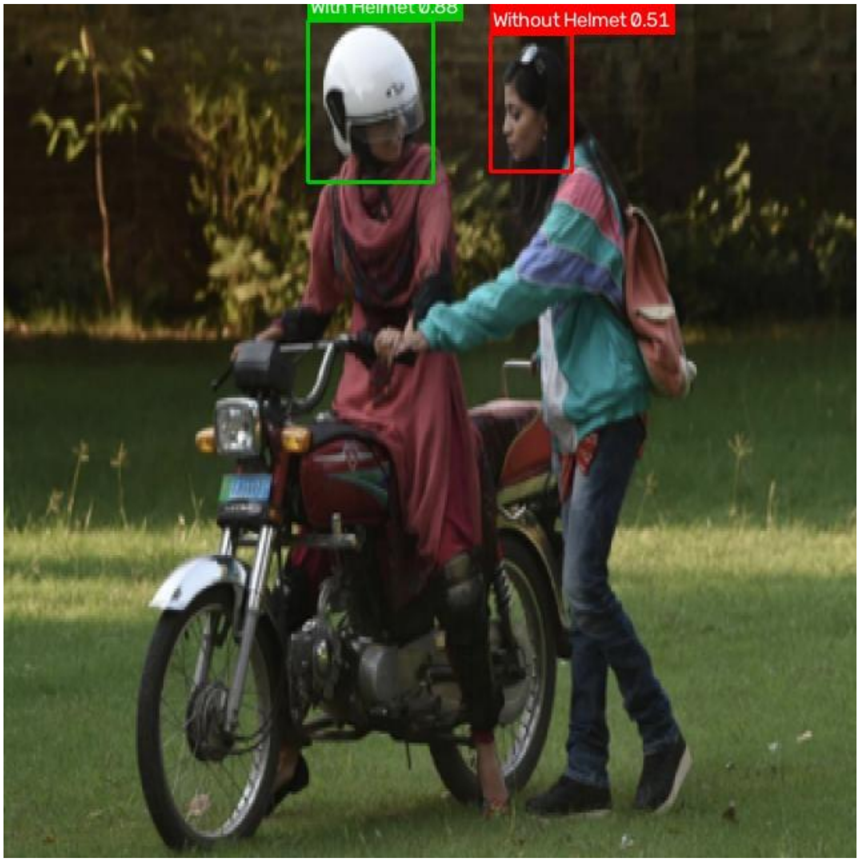
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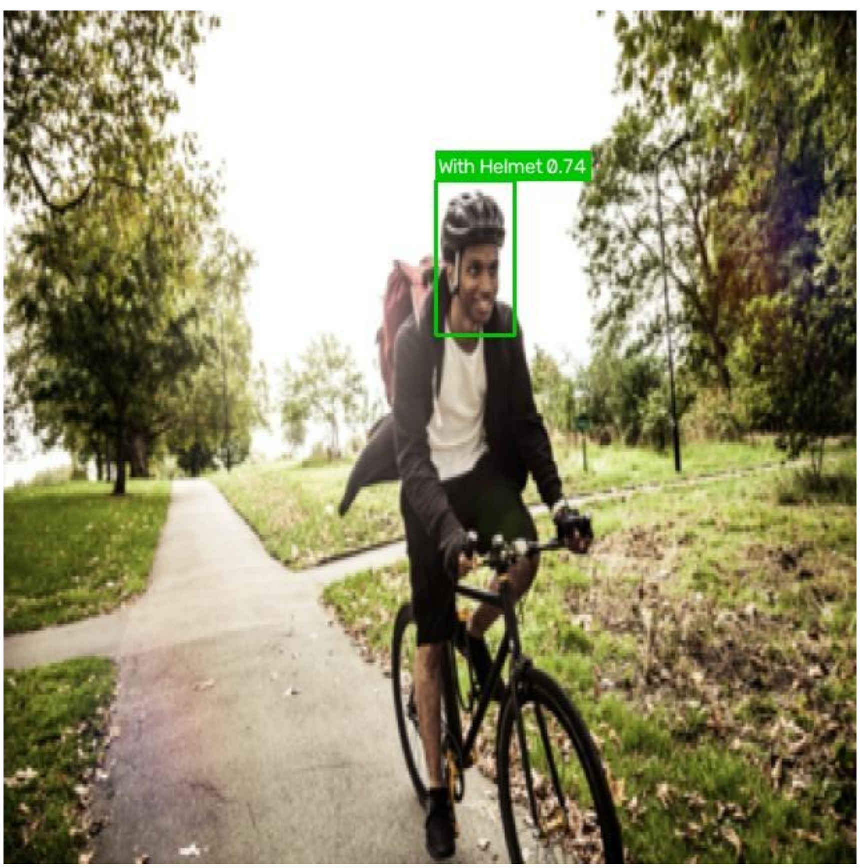
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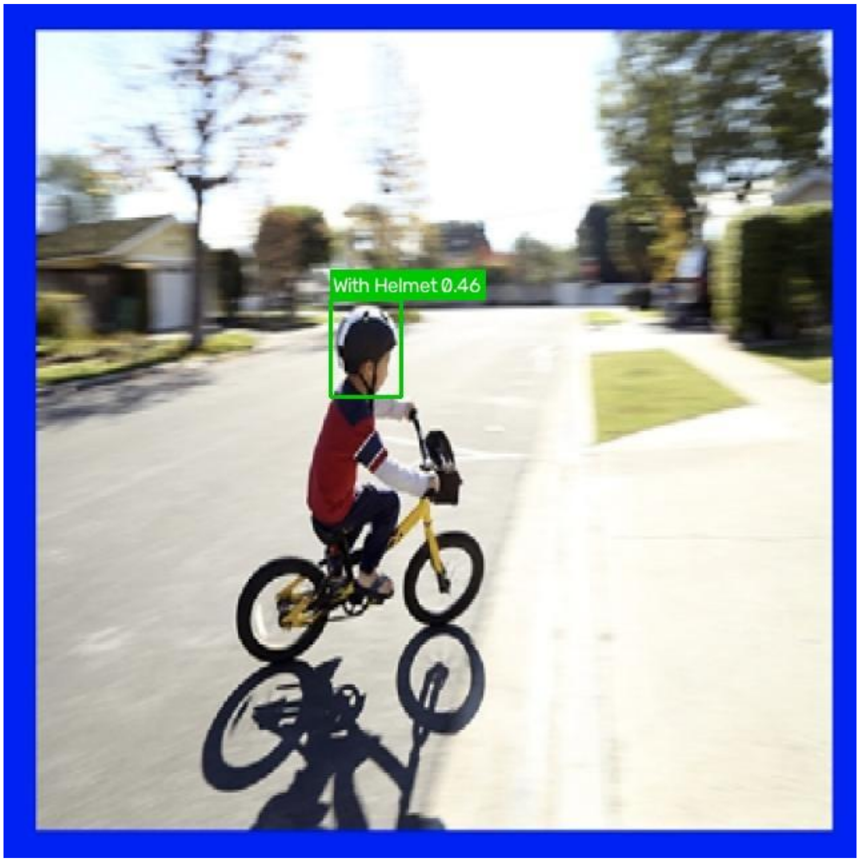
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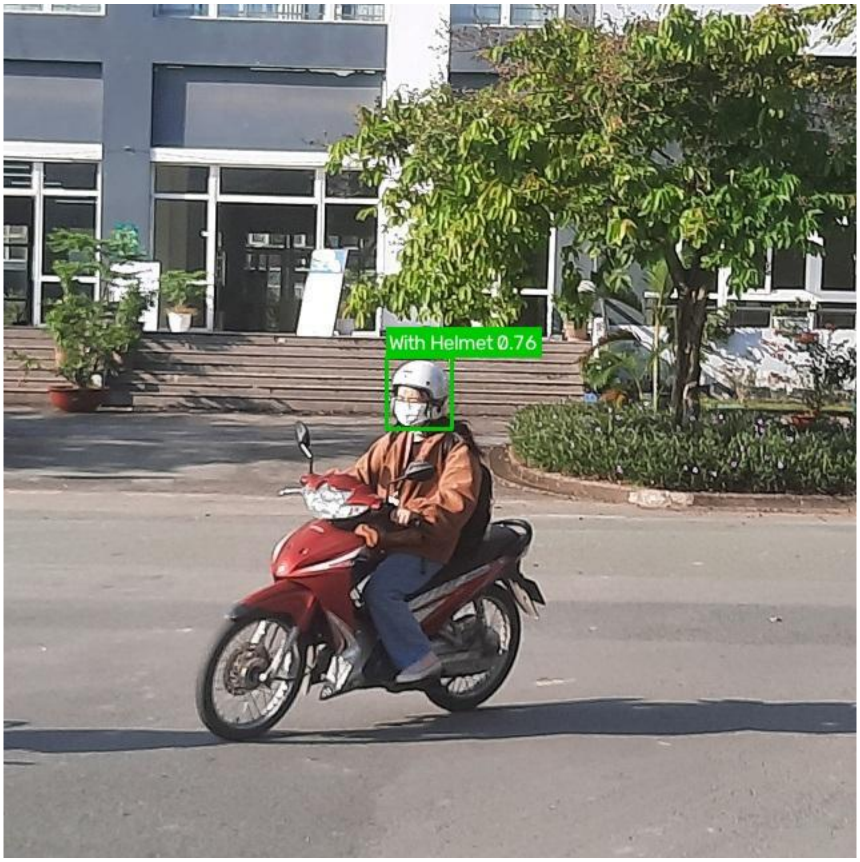
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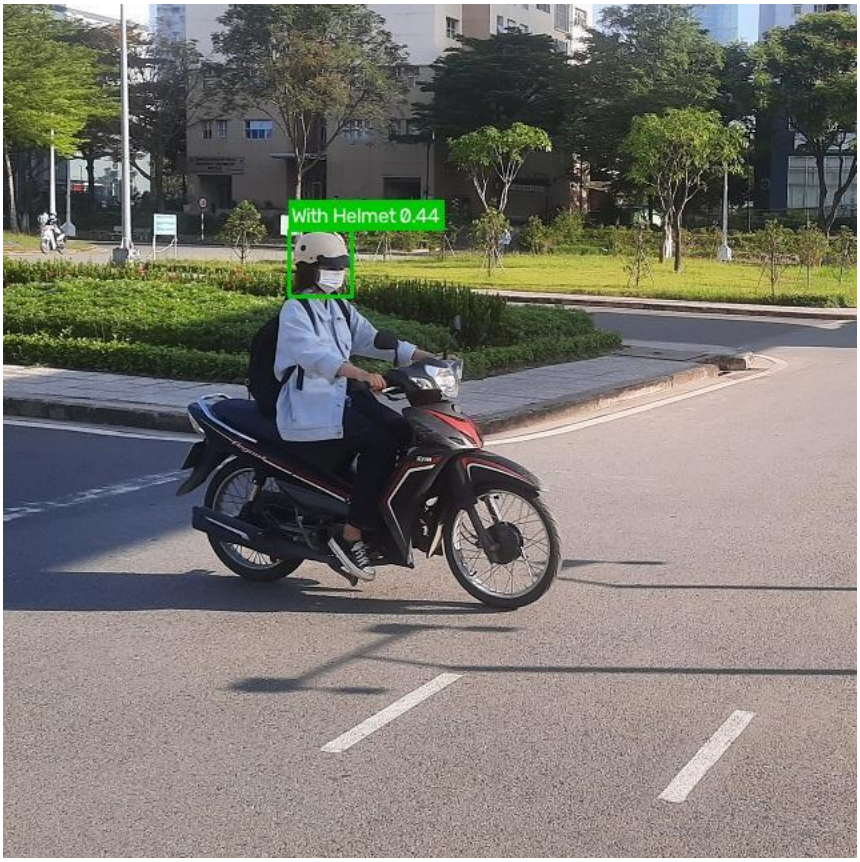
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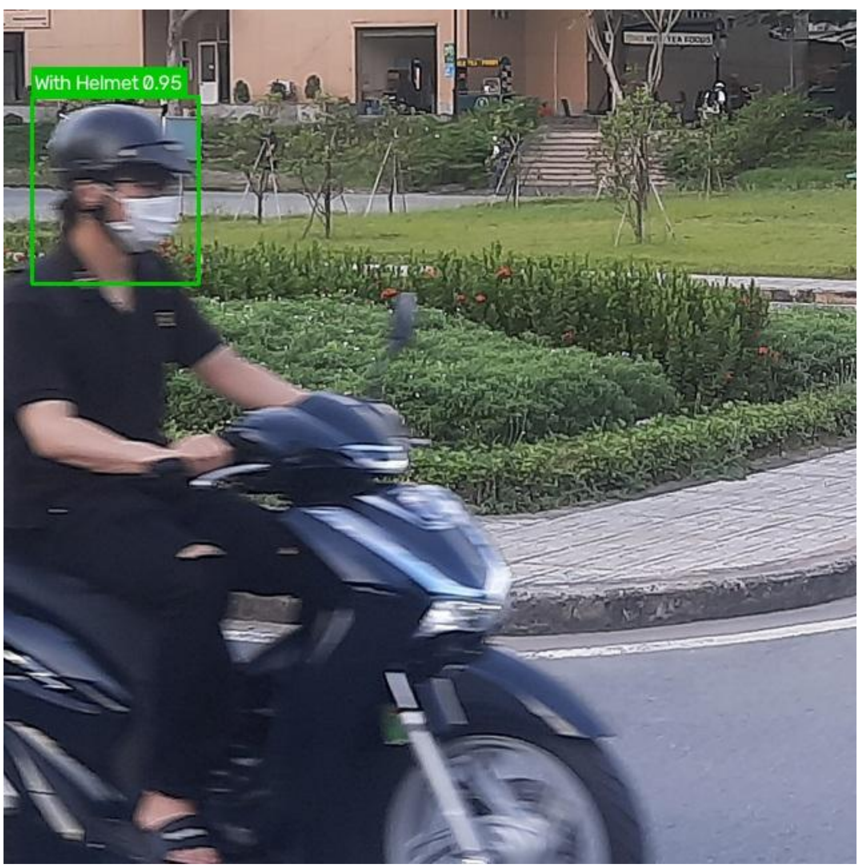
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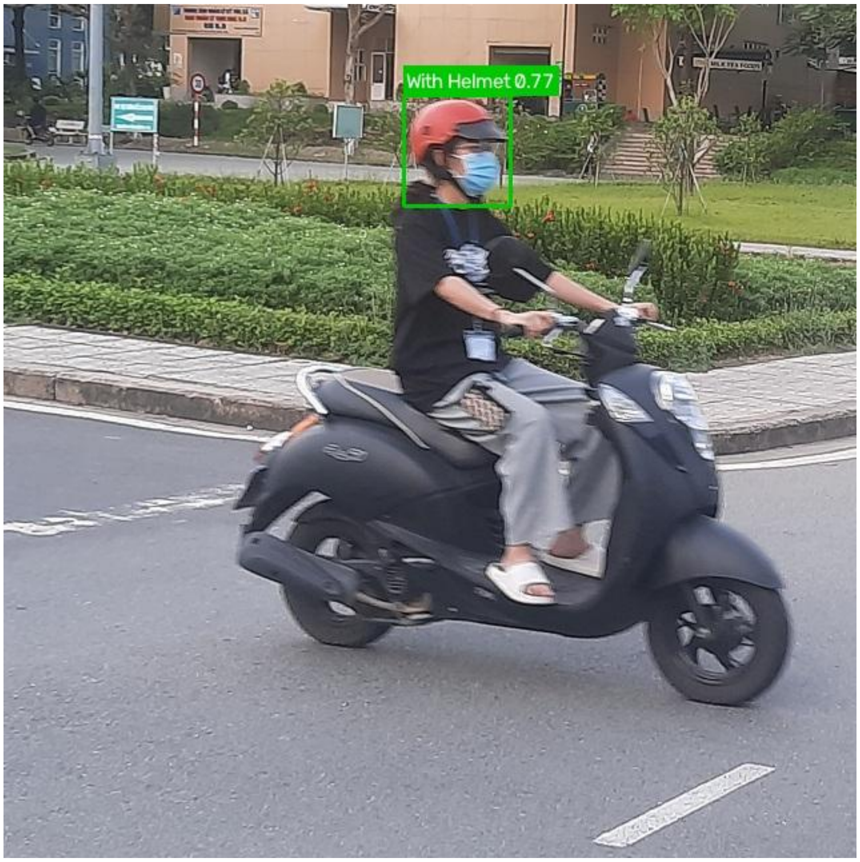
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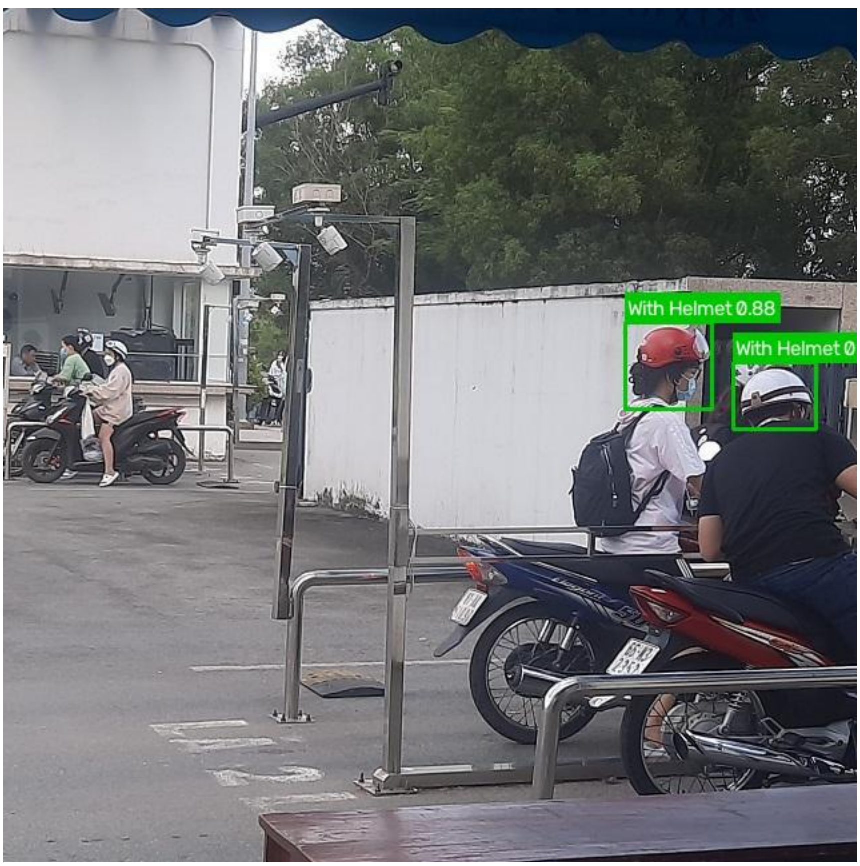
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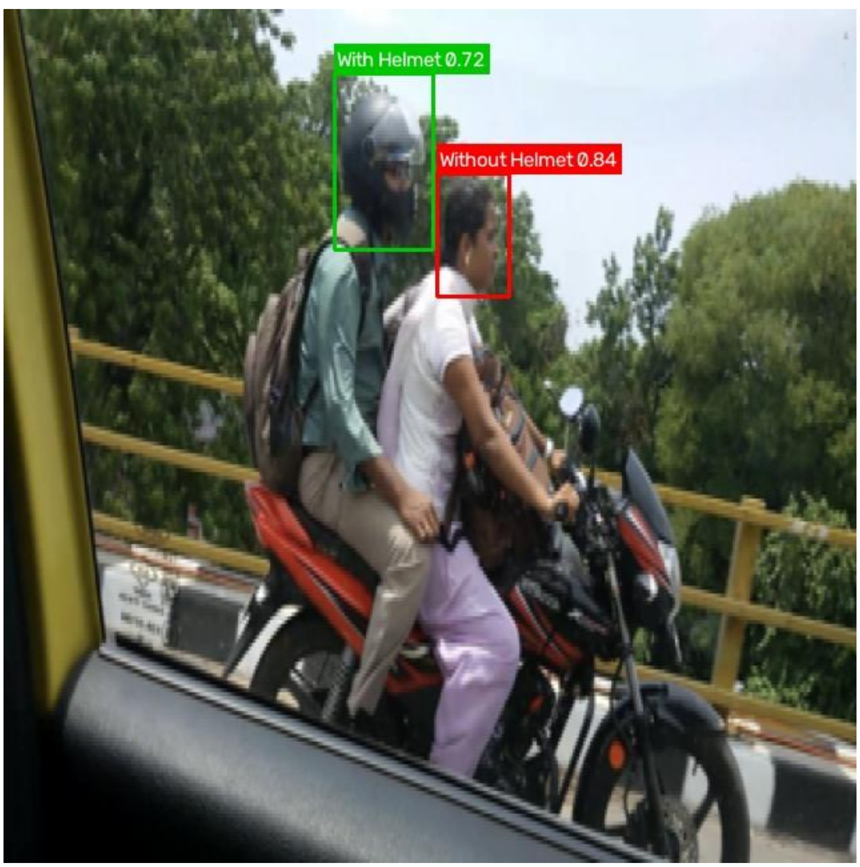
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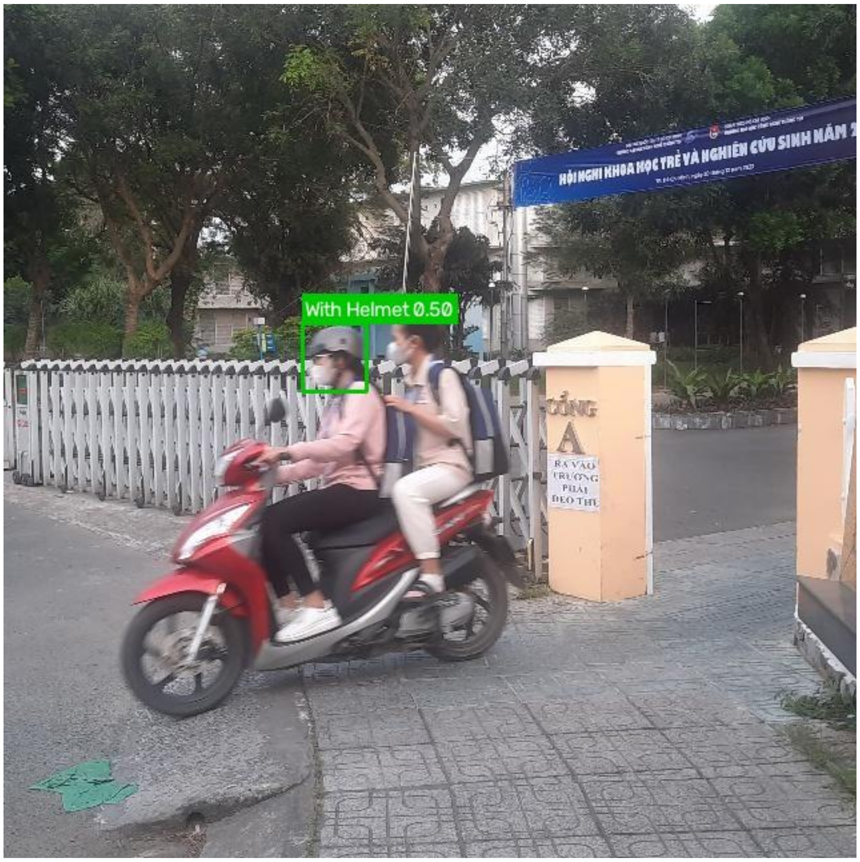
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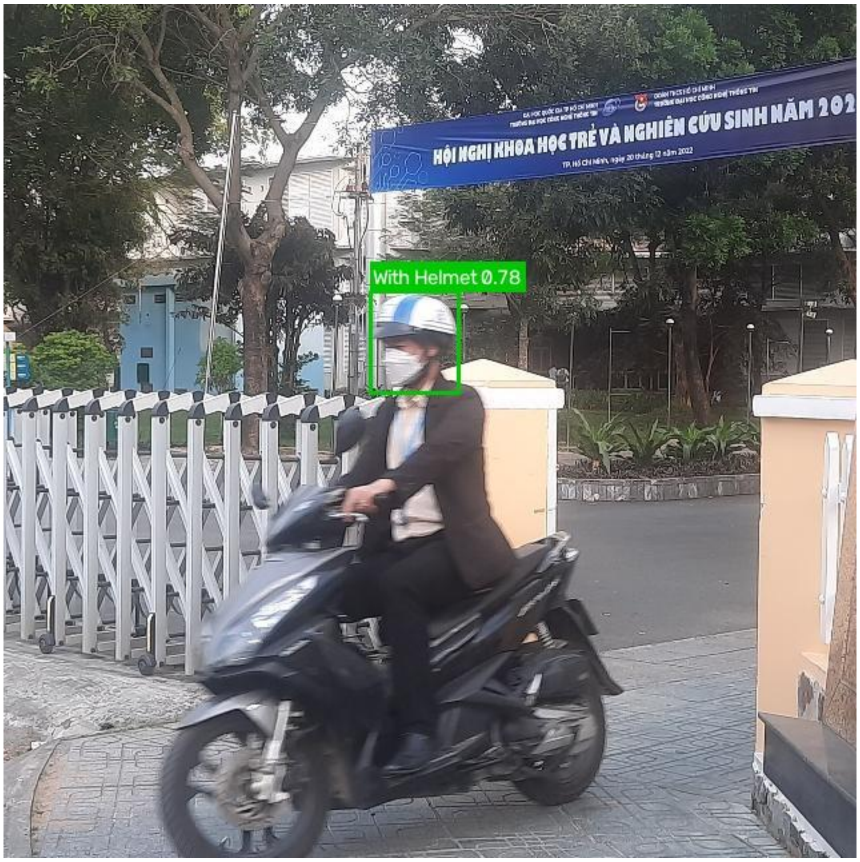
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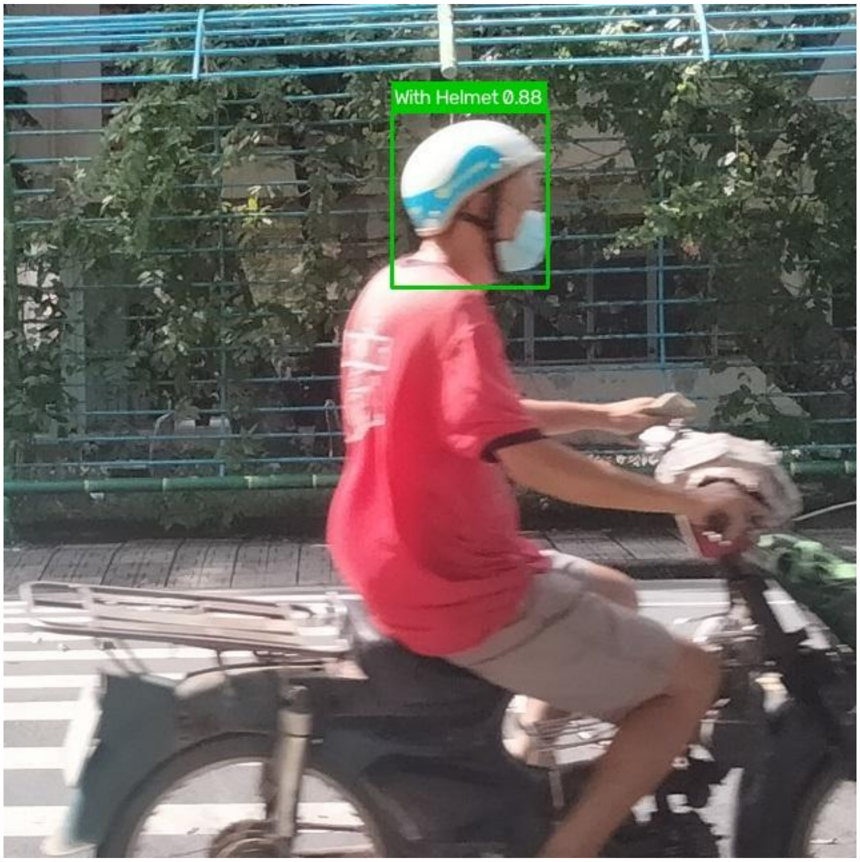
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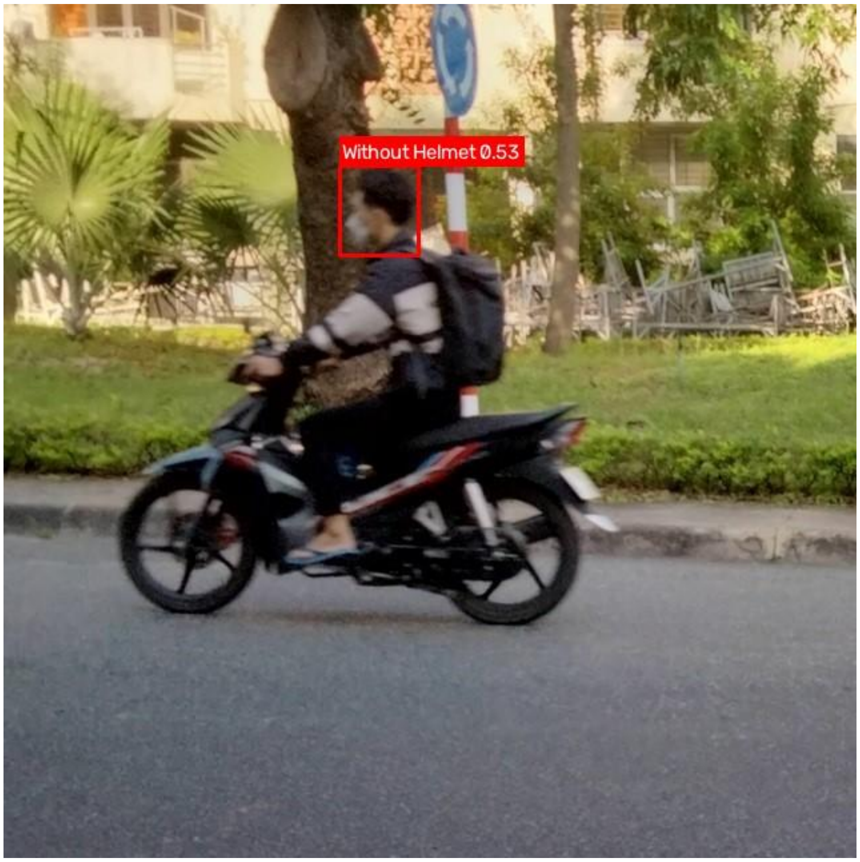
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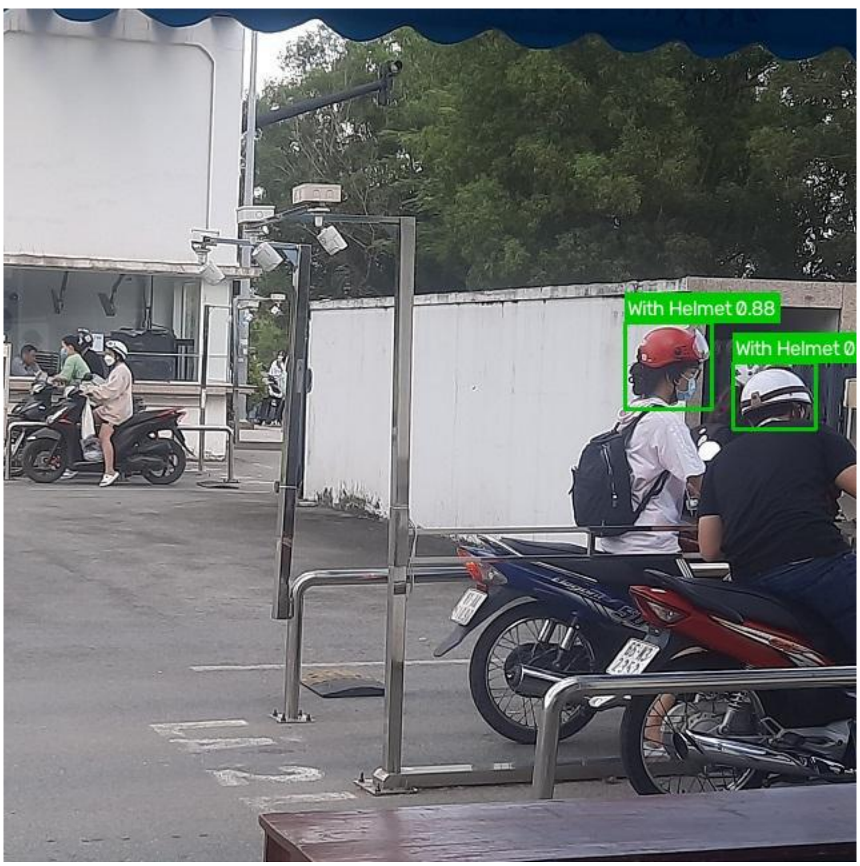
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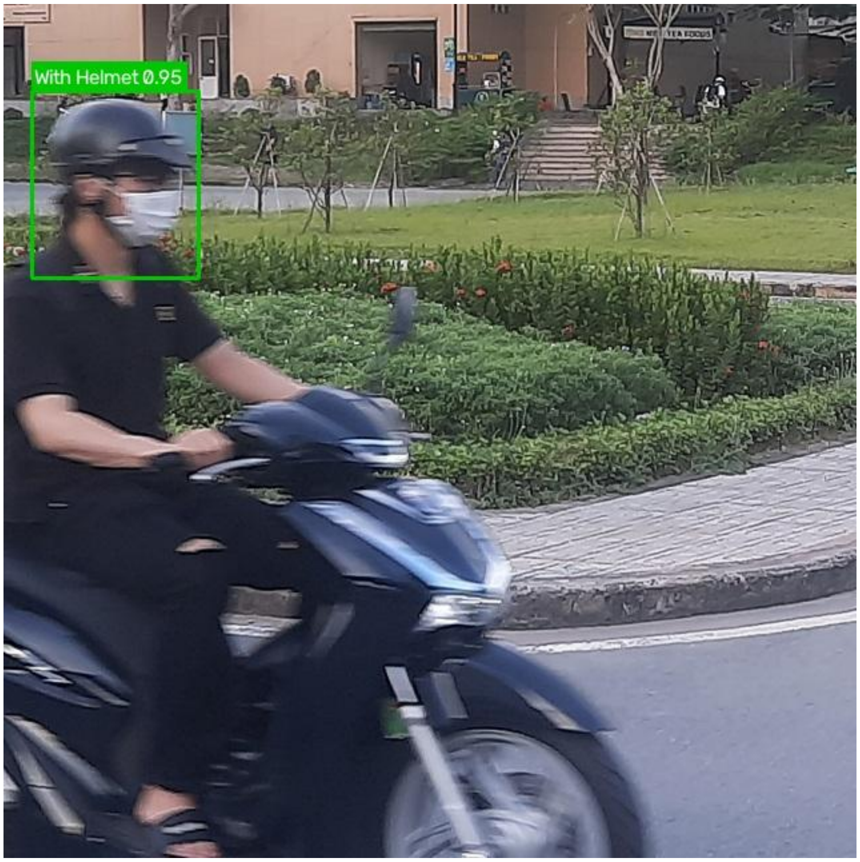
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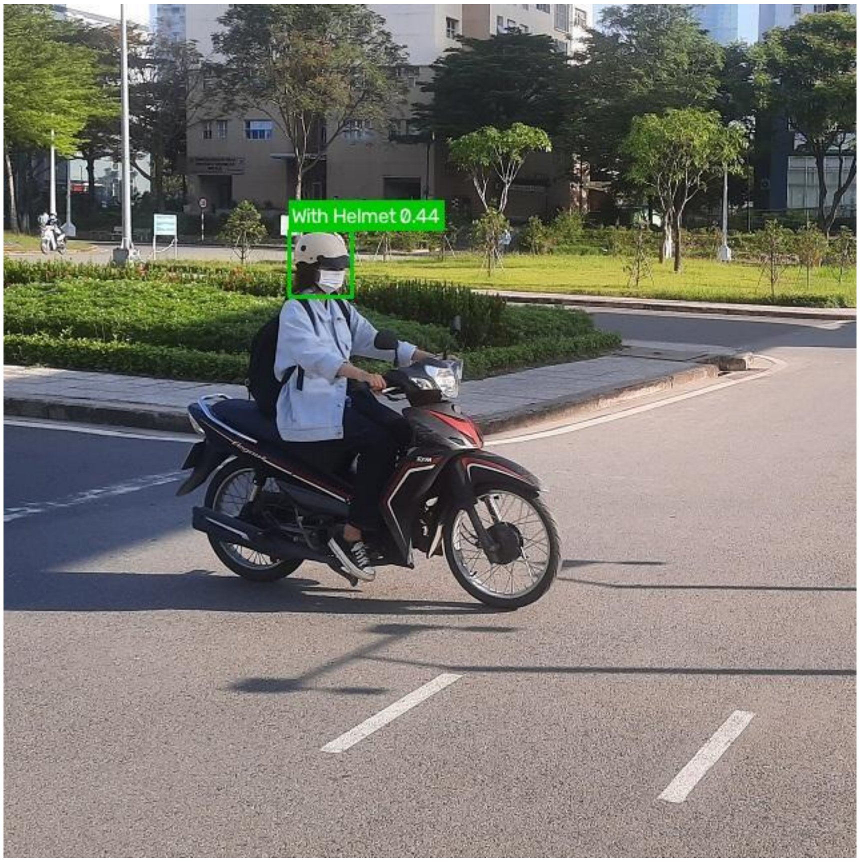
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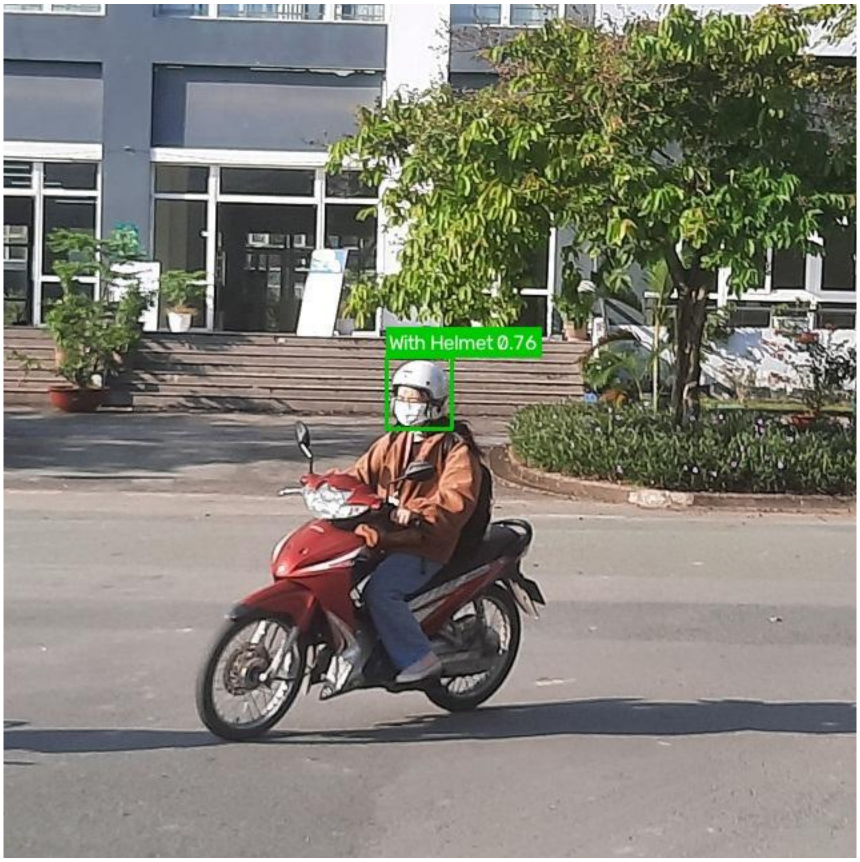
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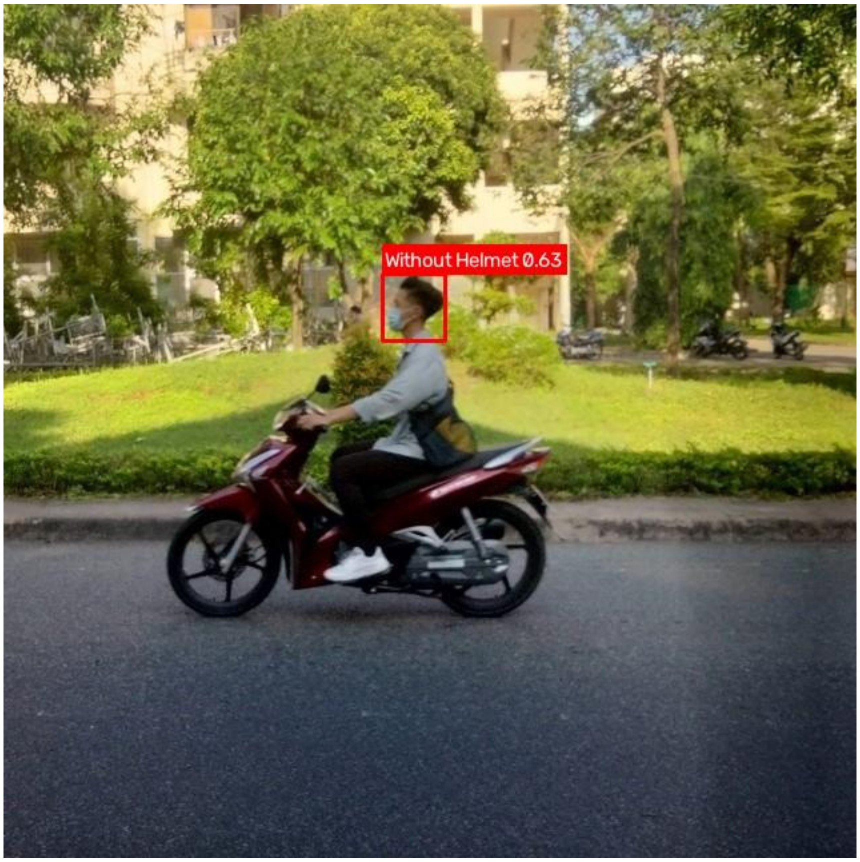
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